

INDEPENDENT PEER-REVIEW PACKET

P02-LM-008

Exact Determinant-Weighted Palm-Tail Transfer

Cold Review Packet 20 of 20

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Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

- 1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
- 2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
- 3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
- 4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	20 of 20
Object ID	P02-LM-008
Object class	Abstract probability interface
Exact statement	If $W_r \geq 0$, $Z_r = E_Q[W_r] \geq c_Z r^2$, and $E_Q[W_r^2] \leq C_W r^4$, then under $dP_r = (W_r/Z_r)dQ_r$, $P_r(E) \leq \sqrt{C_W/c_Z} * Q_r(E)^{(1/2)}$ for every event E.
Internal terminal label	CLOSED - EXACT DETERMINANT-WEIGHTED PALM-TAIL TRANSFER INTERFACE
Terminal date	2026-07-24
Recommended expertise	Probability / change of measure / rare-event bounds

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	One-line Cauchy-Schwarz argument, exact r-power cancellation, and partial-scope independent algebra review.
Prior review record	The unaffected transfer algebra received independent review; an unrelated historical Gaussian-process error is fenced out.
Binding exclusions	Does not prove the normalizer, second moment, base-law event tail, exact-field capture, or P0.2.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.

- Object-specific exclusions: Does not prove the normalizer, second moment, base-law event tail, exact-field capture, or P0.2.

4. Object-specific hostile checks

- Apply Cauchy-Schwarz with the exact same Q_r and W_r used in P_r .
- Check numerator and denominator r powers cancel as claimed.
- Test signed weights, denominator lower/upper reversal, and law mismatch as hostile mutations.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

LCR-CAP-024-v1.0 — P02-LM-008 CLOSED EXACT DETERMINANT-WEIGHTED PALM-TAIL TRANSFER INTERFACE

PEER-REVIEW CAPSULE AND TERMINAL RECORD

Artifact ID: LCR-CAP-024-v1.0

Lemma ID: P02-LM-008

Closure ID: LCR-CLS-023-v1.0

Prepared and adjudicated by: ChatGPT — OpenAI GPT-5.6 Thinking, Lemma Closure Director

Date: 2026-07-24

Class: CAP / CLS / EXACT MEASURE-REWEIGHTING INTERFACE

Authority: terminal lemma record within exact conditional scope under OP-PROT-011

Canonical mathematical impact: NONE

Terminal label: CLOSED — EXACT DETERMINANT-WEIGHTED PALM-TAIL TRANSFER INTERFACE

[[OBJ:P02-LM-008]] [[STATUS:CLOSED-EXACT-CONDITIONAL]] [[CAUCHY-SCHWARZ:EXACT]] [[GT5:NOT-INCLUDED]]
[[NORMALIZER-PREMISE:EXPLICIT]] [[P0.2:OPEN]]

1. READ FIRST

One-sentence result:

If a nonnegative determinant weight has second moment at most $C_W r^4$ and its normalizer is at least $c_Z r^2$, then reweighting any event E from the conditioned law Q_r to the corresponding typed Palm law P_r costs at most a square root:

$$P_r(E) \leq \sqrt{C_W / c_Z} \cdot Q_r(E)^{(1/2)}.$$

Why it matters:

This closes the exact abstract passage from a conditioned-law high-jet or bad-event tail to the determinant-weighted Palm law. It explains why the r^4 weight second moment and r^2 normalizer powers cancel exactly. It is independent of how the conditioned-law event probability is proved.

Primary historical source:

GP-DER-046-v1.0, Palm corollary section.

Drive ID: 1LyYDRVmTl0AKZcKhMDaRMEiWE_HDsmfwac4l2QsFV7c.

Independent review:

CL-AUD-047-v1.0, current corrected version, Section 11.

Drive ID: 1RMDzZH_DBG9MdOY5cz4864shngTn6ShdbdT7K71JgX8.

CL-AUD-047 explicitly independently checked that

$$(C_W r^4)^{(1/2)} / (c_Z r^2)$$

has exact cancellation of all r -powers. Its historical approval of the original GT5 expected-supremum step has been retracted and receives zero terminal credit for P02-LM-007. That unrelated defect does not affect the Cauchy–Schwarz identity frozen here.

Closed prerequisite available in the Drive:

P02-LM-002 — exact uniform r^4 determinant-weight second-moment interface.

Closure: LCR-CAP-011-v1.0.

Drive ID: 1P84b7pNI-z8l9jQrRWW9pP5dNcA4TszKIhMZDFxBlWw.

The normalizer lower bound remains an explicit premise. P02-LM-006 is the open exact-normalizer theorem intended to supply it in the P0.2 application.

Current status:

CLOSED — EXACT DETERMINANT-WEIGHTED PALM-TAIL TRANSFER INTERFACE.

2. FROZEN EXACT STATEMENT

Let $(\Omega, \mathcal{F}, Q_r)$ be a probability space for each $0 < r \leq r_0$. Let

$W_r: \Omega \rightarrow [0, \infty)$

be measurable with

$0 < Z_r := E_{Q_r}[W_r] < \infty$.

Define the W_r -reweighted probability law P_r by

$$P_r(E) = E_{Q_r}[W_r \mathbf{1}_E] / Z_r$$

for every measurable event E .

Assume constants $c_Z > 0$ and $C_W < \infty$, independent of r , such that

$$Z_r \geq c_Z r^2$$

and

$$E_{Q_r}[W_r^2] \leq C_W r^4$$

for every $0 < r \leq r_0$.

THEOREM P02-LM-008.

For every measurable event E and every $0 < r \leq r_0$,

$$P_r(E)$$

$$\leq \sqrt{C_W / c_Z} \cdot Q_r(E)^{(1/2)}.$$

Consequences:

A. Polynomial tail transfer.

If

$$Q_r(E_r) \leq C_E r^\beta$$

for some $\beta \geq 0$ and all sufficiently small r , then

$$P_r(E_r) \leq \sqrt{C_W C_E} / c_Z \cdot r^{(\beta/2)}.$$

B. Sub-Gaussian small- r transfer.

If

$$Q_r(E_r) \leq C_E \exp(-c_E r^2),$$

then

$$P_r(E_r) \leq \sqrt{C_W C_E} / c_Z \cdot \exp[-(c_E/2)r^2].$$

C. Super-polynomial transfer.

If $Q_r(E_r) = o(r^m)$ for every $m > 0$, then $P_r(E_r) = o(r^p)$ for every $p > 0$.

No independence assumption, Gaussian assumption, or relationship between E_r and the weight coordinates is required.

3. EXACT PROOF

By Cauchy–Schwarz under Q_r ,

$$\begin{aligned} E_{\{Q_r\}}[W_r 1_E] &\leq E_{\{Q_r\}}[W_r^2]^{(1/2)} \\ E_{\{Q_r\}}[1_E^2]^{(1/2)}. \end{aligned}$$

Since $1_E^2 = 1_E$,

$$E_{\{Q_r\}}[1_E^2]^{(1/2)} = Q_r(E)^{(1/2)}.$$

Using the second-moment premise,

$$E_{\{Q_r\}}[W_r 1_E] \leq \sqrt{C_W} r^{(1/2)} Q_r(E)^{(1/2)}.$$

Dividing by the lower normalizer bound gives

$$\begin{aligned}
& P_r(E) \\
&= E_{\{Q_r\}}[W_{r1_E}/Z_r] \\
&\leq \sqrt{(C_W)r^2} \cdot Q_r(E)^{1/2} / (c_Z r^2) \\
&= \sqrt{(C_W)/c_Z} \cdot Q_r(E)^{1/2}.
\end{aligned}$$

This proves the theorem.

The three corollaries follow by direct substitution.

For the super-polynomial statement, fix $p > 0$. Apply the theorem using any conditioned-law rate $o(r^{(2p)})$; the square root is $o(r^p)$.

4. PROOF-OBLIGATION LEDGER

PT1 — P_r is a probability law.

Status: PASS under the exact premise $0 < Z_r < \infty$ and $W_r \geq 0$.

PT2 — Cauchy–Schwarz applicability.

Status: PASS because W_r is square-integrable by the stated premise and 1_E is bounded.

PT3 — Event dependence.

Status: PASS. E may depend arbitrarily on all underlying coordinates. No independence is used.

PT4 — Numerator power.

Status: PASS. The square root of $C_W r^4$ is $\sqrt{C_W} r^2$.

PT5 — Denominator direction.

Status: PASS. An upper bound on the ratio uses the lower bound $Z_r \geq c_Z r^2$.

PT6 — Exact r -power cancellation.

Status: PASS. $r^2/r^2 = 1$.

PT7 — Exponential-rate halving.

Status: PASS. Taking the square root of $\exp(-c_E r^2)$ gives $\exp[-(c_E/2)r^2]$.

PT8 — Super-polynomial preservation.

Status: PASS because the input estimate is available at every power.

PT9 — Same-law requirement.

Status: PASS and explicit. The numerator, second moment, normalizer, and event probability must all be taken under the identical Q_r and W_r .

PT10 — No hidden finite-Q4/exact-field substitution.

Status: PASS at the abstract level. A concrete application must separately crosswalk its exact law and weight.

5. INDEPENDENT REVIEW AND LINEAGE

CL-AUD-047-v1.0 was produced by the Anthropic/Claude line and independently reviewed the V2 proof architecture.

Its current correction banner retracts the historical false expected-supremum inequality and explicitly denies terminal credit for the repaired P02-LM-007 theorem.

The current document preserves an unaffected, separately checkable finding in Section 11:

- - the Cauchy–Schwarz reweighting step is correct;
- - the exact factor $(C_{Wr}^4)^{(1/2)}/(c_{Zr}^2)$ has no remaining power of r ;
- - the DM2 dependence is properly separated from the conditioned-law tail.

Review classification for P02-LM-008:

ORGANIZATIONALLY DISTINCT / PARTIAL-SCOPE ALGEBRAIC REVIEW / ORIGINAL ARGUMENT VISIBLE.

The review counts once for this exact extracted interface. It does not transfer to P02-LM-007 or to any premise of P02-LM-008.

6. ADVERSARIAL MUTATIONS

M1 — Use an upper normalizer bound.

Mutation:

$$Z_r \leq C_{Zr}^2.$$

Result:

Does not upper-bound the ratio. This mutation kills the theorem’s conclusion.

M2 — Replace the second moment by a first moment.

A bound $E[W_r] \leq C_{r^2}$ alone does not give the same event-uniform square-root transfer through Cauchy–Schwarz. Additional structure would be needed.

M3 — Let the second moment scale as r^γ .

The general ratio power becomes $r^{(\gamma/2-2)}$. The scale-free conclusion is special to $\gamma=4$.

M4 — Compute the event probability under another law.

Cauchy–Schwarz requires $Q_r(E)$, not an adjacent finite-Q4, unconditioned, or differently conditioned event probability.

M5 — Assume independence between W_r and E .

Unnecessary. The theorem is stronger because it does not use independence.

M6 — Use a signed weight.

The ratio may fail to define a probability law. Nonnegativity and positive finite normalizer are load-bearing.

M7 — Infer a conditioned-law tail from the theorem.

The implication is one-way: conditioned-law tail plus moment/normalizer premises gives Palm tail. The theorem does not prove $Q_r(E)$.

M8 — Import the retracted GT5 step.

Irrelevant and prohibited. P02-LM-008 accepts any valid event-probability premise, regardless of its source.

7. EDGE CASES

$Q_r(E)=0$:

Then $1_E=0$ Q_r -almost surely, the weighted numerator is zero, and $P_r(E)=0$.

$Q_r(E)=1$:

The theorem gives $P_r(E) \leq \sqrt{C_W}/c_Z$. This constant may exceed one; the true probability also satisfies the trivial bound one. The theorem remains correct but not necessarily sharp.

$W_r=0$ on E :

Then $P_r(E)=0$, consistent with the bound.

Type-boundary or singular-weight events:

Allowed. No continuity of the weight is required.

Different r -dependent events:

Allowed. The proof is pointwise in r .

8. EXCLUSIONS

This closed lemma does not establish:

- - the second-moment premise for a concrete weight;
- - the normalizer lower bound;
- - any conditioned-law tail;

- P02-LM-006;**- P02-LM-007;**

- - exact-field capture;
- - a boundary-layer estimate;

- P0.2;

- - any numerical Palm probability;
- - theorem promotion or release.

P02-LM-002 supplies the second-moment interface for the exact endpoint determinant weight under a uniform eighth-moment premise. P02-LM-005 supplies that exact-law moment premise. P02-LM-006 is intended to supply the exact normalizer bound. P02-LM-007 is intended to supply one conditioned-law tail.

9. CLOSURE DECISION

Closure ID: LCR-CLS-023-v1.0

Lemma ID: P02-LM-008

Date: 2026-07-24

Terminal label:

CLOSED — EXACT DETERMINANT-WEIGHTED PALM-TAIL TRANSFER INTERFACE.

Decisive evidence:

- - exact one-line Cauchy–Schwarz proof;
- - explicit same-law and denominator-direction conditions;
- - organizationally distinct partial-scope algebraic review;
- - historical GT5 error quarantined from the extracted interface;
- - complete premise and downstream scope fence.

OP-PROT-011 independent-eyes condition is satisfied for this exact interface by the unaffected CL-AUD-047 Section 11 review.

10. DOWNSTREAM USAGE

Authorized exact use:

If a concrete exact-law package proves

$$\begin{aligned} E_{\{Q_r\}}[W_r^2] &\leq C_W r^4, \\ Z_r &\geq c_Z r^2, \\ Q_r(E_r) &\leq T_r, \end{aligned}$$

then it may cite P02-LM-008 to conclude

$$P_r(E_r) \leq \sqrt{C_W}/c_Z \cdot \sqrt{T_r}.$$

For the P0.2 chain:

- - P02-LM-002 + P02-LM-005 provide the exact-law second-moment bound;
- - P02-LM-006, if independently approved, provides the normalizer lower bound;
- - P02-LM-007, if independently approved, provides the conditioned H5 tail;
- - P02-LM-008 then supplies the determinant-weighted H5 tail without further independence assumptions.

Every application must verify that all four objects use the identical Q_r and W_r .

11. REOPENING CONDITIONS

Reopen if:

- - the reweighted-law definition changes;
- - W_r is allowed to be signed;
- - the normalizer or second moment belongs to a different law/weight;
- - the historical independent-review record is shown not to have checked the exact Cauchy–Schwarz cancellation;
- - an algebraic error is found in the ratio direction or power cancellation.

Do not reopen merely because a sharper Hölder exponent or event-specific bound is found.

12. MANIFEST

Historical primary derivation:

- - GP-DER-046-v1.0
- - Drive ID 1LyYDRVmTl0AKZcKhmdARMEiWE_HDsmfwac4l2QsFV7c
- - Role: original Palm corollary and broader GT5 proof
- - Status for this lemma: controlling algebraic source only

Independent partial-scope review:

- - CL-AUD-047-v1.0, corrected current version
- - Drive ID 1RMDzZH_DBG9MdOY5cz4864shngTn6ShdbdT7K71JgX8
- - Role: independent Cauchy–Schwarz and power-cancellation check
- - Status: qualifying for P02-LM-008 only; nonqualifying for repaired GT5 theorem

Closed second-moment input:

- - P02-LM-002 / LCR-CAP-011-v1.0
- - Drive ID 1P84b7pNI-z8l9jQrRWW9pP5dNcA4TszKIhMZDFxBIWw

Closure capsule:

- - LCR-CAP-024-v1.0 / LCR-CLS-023-v1.0
- - Drive ID 1tQvVF3Q2kTNf1lh_XSW3CB_DfOhcLS6vaWPGavUYfio

13. FINAL CHECKLIST

Exact statement: PASS.

Measurability and integrability premises: PASS.

Proof complete: PASS.

Same-law requirement explicit: PASS.

Denominator direction: PASS.

Power cancellation: PASS.

Independent partial-scope review: PASS.
Historical false GT5 step quarantined: PASS.
Premise/exclusion fence: PASS.
P0.2 separation: PASS.

FINAL VERDICT:

APPROVE AND CLOSE P02-LM-008 WITHIN THE EXACT CONDITIONAL DETERMINANT-WEIGHTED REWEIGHTING SCOPE.

END LCR-CAP-024-v1.0

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT DETERMINANT-WEIGHTED PALM-TAIL TRANSFER INTERFACE
Closure date	2026-07-24
Evidence basis	One-line Cauchy-Schwarz argument, exact r-power cancellation, and partial-scope independent algebra review.
Independent review already recorded	The unaffected transfer algebra received independent review; an unrelated historical Gaussian-process error is fenced out.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- Does not prove the normalizer, second moment, base-law event tail, exact-field capture, or P0.2.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____
Reproduction artifacts	Code/log/hash/location: _____

Signature and date _____

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: https://docs.google.com/document/d/1tQvVF3Q2kTNf1Ih_XSW3CB_DfOhcLS6vaWPGavUYfio/edit

Closure / terminal record:

https://docs.google.com/document/d/1tQvVF3Q2kTNf1Ih_XSW3CB_DfOhcLS6vaWPGavUYfio/edit

Local source snapshot used to construct this packet: LCR-CAP-024-v1.0 — P02-LM-008 Closed Exact Determinant-Weighted Palm-Tail Transfer Interface

Local source snapshot SHA-256 (DOCX export, transport checksum only):
282b68c79116d435ead138ba6b66fd3195c9c00ade171f5be2cd2839e1841fe0

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.